

Why hash algorithms have many different digest size variants?

Asked 1 year, 3 months ago Modified 1 year, 3 months ago Viewed 473 times

- 1- Why SHA2 has SHA224, SHA256, SHA384 and SHA512 variants?
- 2- Can we say SHA512 more secure than SHA256?
- 3- Symmetric ciphers use at most 2^{256} security level and I saw on the internet people saying how impossible to reach that then Why SHA2 has a SHA 512 variant? Do we need 2^{512} security level? Isn't it overkill?
- 4- Is SHA 512 meant for post quantum world? If it's meant for post quantum world, Why does it not popular? because AES256 already in use by encrypted messengers, password managers.. etc. but I've seen their security whitepapers when it comes to SHA2 most of them use SHA256 instead of SHA512.
- 5- so if we don't need 512 bit security, Why all the state of the art hash algorithms(e.g. BLAKE2, Skein, Grostl, JH, Keccak) have 512 bit variants?
- 6- Let's say there's a key derivation function that uses Skein 256 as a underlying hash function, Would it be still secure in a post quantum world? or Would it be better to use Skein 512? to ask it another way, Why Argon2 settled on using BLAKE2b 512 instead of BLAKE2s 256?

sha-256

Share Edit Follow Flag

asked Sep 29, 2023 at 12:21

user111059

- 2 Not quite an answer, but there's an incorrect assumption in the question, specifically in 3) and 5): An n -bit hash function only provides $n/2$ -bit security versus collision attacks. So if you wanted a 256-bit security level (to pair with e.g. AES-256, or an appropriately sized elliptic curve or RSA modulus) you'd have to use a 512-bit hash function. – Morrolan Sep 29, 2023 at 12:57
- What a broad question! Even the first part has long answer ; [SHA-224 Purpose](#) – kelalaka Sep 29, 2023 at 13:56
- [Why SHA-512/256 when we already have SHA-384?](#) – kelalaka Sep 29, 2023 at 13:57
- [Difference between SHA 512, SHA 512 Half, SHA 256](#) – kelalaka Sep 29, 2023 at 13:58
- SHA-512 designed for 64-bit processors. – kelalaka Sep 29, 2023 at 14:00

1 Answer

Sorted by: Highest score (default)

- 3 1. The main reasons for that are efficiency (shorter bits generally need lesser computational power) and security (more bits generally imply higher security). A third reason regarding the real world is: Systems are build for a specific bit length, which implies, that variations may be needed. This is the general statement. For a more detailed explanation regarding SHA see [this](#) and [this](#) post.
2. In general, SHA-512 is considered more secure than SHA-256 in the context of cryptographic hash functions. The primary reason for this is the difference in their output sizes and the resistance to collision attacks. But there are not only relevant aspects in terms of security, the technical side (specifically interesting for use cases) is also different. SHA256 works with 32 bit, SHA512 bit blocks. Depending on the hardware and the use case, this can also influence the efficiency. See [this](#) for more details on efficiency, [this](#) for security and [this](#) for a use case.
3. Empirically gained practical experience leaves some room for discussion, which is not necessarily scientifically based. The problem is that it is difficult to make precise statements for practice. Every system, every use case, etc. is different. My rule of thumb is that 128 bit (symmetric scheme and 256 bit hash functions) can be considered secure for personal use. But 256 or even 512 may be relevant for other usecases, e.g. military use. But regarding my rule of thumb you could call 512 bit security overkill for personal use.
4. The problem with hash functions and symmetric cryptography in the post quantum domain is the Grover algorithm. Simplified, it delivers an attack that lowers the bit security by a logarithmic factor. Therefore, symmetric cryptography is not affected so much, since the key length can simply be doubled. For hash functions there is an algorithm from Brassard, which breaks the whole thing even more efficiently. I.e. a 256 bit hash function is not PQ secure to the same extent as a 256 symmetric encryption. Instead of the 2nd root, the 3rd root is added. I.e. the bit number must be selected still more highly, why one can say that SHA512 is rather for PQ (but not only, evenly because of the already mentioned other design decisions). For more see [this](#)
5. There is a saying: Better to have and not need than to need and not have.
6. As mentioned above: It depends.

Share Edit Follow Flag

edited Oct 2, 2023 at 11:35

answered Sep 29, 2023 at 13:43

Titanlord
2,457 12 34

- thanks a lot for the detailed explanation, got it! so SHA 512 has a 256 bit security level and overkill to use in today's world, right? compare to SHA256. Can we say Argon2 developers considered quantum attacks as a future threat and they did not want to bother with 10 rounds BLAKE2s 256 and used higher security margin variant 12 rounds BLAKE2b 512? Is that the reason? – user111059 Sep 29, 2023 at 14:10
- @adler See [this post](#) for a more detailed comparison of SHA512 and Argon2 :D – Titanlord Oct 2, 2023 at 11:21
- @kelalaka I hope it is better now :D – Titanlord Oct 2, 2023 at 11:39
- One historical justification for SHA-224 is that it's security level (against collision) is 112 bit, which matches the security level of 2-keys TDES against brute force, the security level of 3-keys TDES under some hypothesis, and a conservative evaluation of the security level of RSA-2048. For SHA-384, that's to match the 192-bit security level which is one of three conventional ones (128/192/256) for US unclassified government use, and a 384-bit hash is convenient for ECDSA with P-384 aka [secp384r1](#), designed for 192-bit security level. – fgrieu Oct 2, 2023 at 11:46